CLYC Full-Energy Peak Efficiency for the DCS-Monitor

CLYC2 Distance Comparison, Peak Integration, Detector Resolution, and Flush-Source Response

DCS-Monitor simulation notes

November 3, 2026

Abstract

This report compares simulated and measured full-energy peak efficiencies for the CLYC2 detector at nominal source-to-detector distances of 200 mm, 300 mm, and 400 mm. The simulated efficiency is obtained from isotropic Geant4 runs with 10^7 gammas per point and a $\pm 2\,\%$ total-energy window. The measured efficiency is obtained from the 2026-10-27 CLYC2 spectra after subtracting the 600 s environmental background and fitting each peak with a Gaussian plus linear background over ± 1.5 FWHM. No external efficiency table is used for the distance comparison. At 661.64 keV the simulation/measurement ratio is 0.98–1.03; at 1332.51 keV it is 0.92–1.11. At 1173.23 keV the simulation is 19–36% larger than the measurement. The 1173.23 keV peak sits on the Compton continuum from the 1332.51 keV line; the Gaussian peak area has a 10 % background-model uncertainty, but the excess is larger. The measured 1173 keV/1332 keV ratio is 1.00–1.04, compared with 1.18 in the simulation. Cascade summing is negligible at these distances, so the line-dependent excess points to the Co-60 source geometry or the detector response model. Separately, the 2 cm calibration file is compared with a radioactive-decay simulation in which the source puck is flush with the detector front face. The flush simulation agrees at 46.54 keV, 661.64 keV, and 1408.01 keV and, after pulse clustering, at 59.54 keV; the 121.78 keV, 344.28 keV, and ^{60}Co lines remain high. The detector resolution is $R(E) = (123.9 \pm 0.4)/\sqrt{E/\mathrm{keV}}\,\%$ (FWHM $= 1.239\sqrt{E/\mathrm{keV}}$ keV), fitted to the 661.64 keV and 1332.51 keV lines.

1 Geometry and Solid Angle Efficiency Bound

The detector model uses a cylindrical CLYC crystal of diameter 25 mm and length 25 mm (radius $r = 12.5$ mm, density $3.31\,\mathrm{g\,cm^{-3}}$) placed inside a stepped aluminium casing (`GeometryCLYC.cc`). The front face of the crystal sits 2 mm behind the front surface of the detector casing ($z_{\mathrm{recess}} = 2$ mm), consisting of a 0.5 mm aluminium entrance window and a 1.5 mm Teflon reflector cap. Behind the 29 mm snout, the wider detector body and PMT are surrounded by a 0.64 mm-thick MuMetal magnetic shield. The calibration sources are modeled as isotropic point sources at the centre of an epoxy puck of radius 10 mm, thickness 2 mm, and density $1.16\,\mathrm{g\,cm^{-3}}$ ($H_{32}C_{15}N_2O_4$).

For a nominal source-to-detector distance d_{nom}, the true distance between the source and the crystal front face is $d = d_{\mathrm{nom}} + 2$ mm. Figure 1 shows the axial geometry and the solid-angle cone.

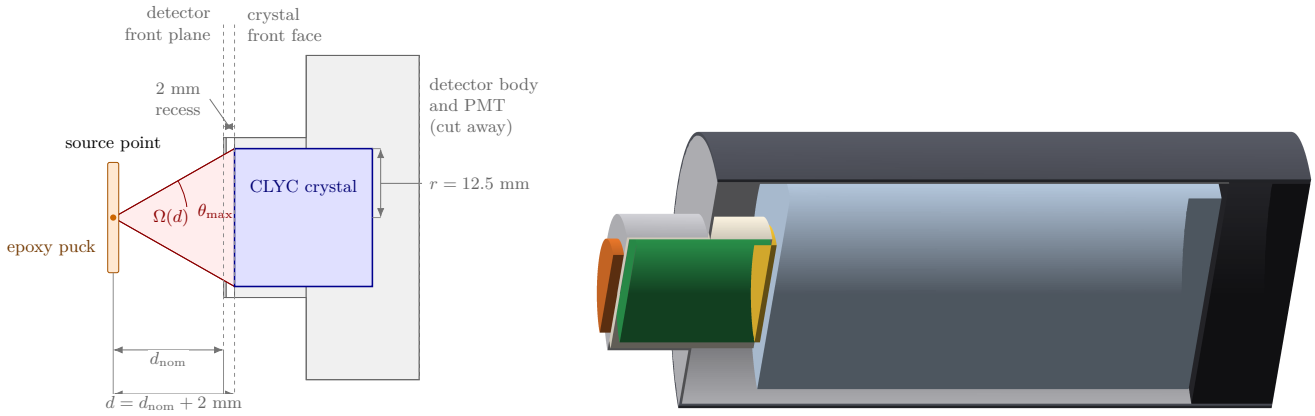
1.1 Solid angle and geometric acceptance

For an isotropic point source emitting into 4π, the solid angle subtended by the circular crystal entrance face of radius r at distance d along the detector axis is

$$\Omega(d) = 2\pi \left(1 - \frac{d}{\sqrt{d^2 + r^2}}\right), \tag{1}$$

and the geometric acceptance is

$$\epsilon_\Omega(d) = \frac{\Omega(d)}{4\pi} = \frac{1}{2}\left(1 - \frac{d}{\sqrt{d^2 + r^2}}\right) \times 100\%. \tag{2}$$



(a) Schematic axial section (not to scale).

(b) Geant4 cutaway half-section.

Figure 1: Axial geometry of the source and detector. Left: the point source is at the centre of the 2 mm epoxy puck; the detector front plane is at d_{nom} and the crystal front face is 2 mm farther, at $d = d_{\text{nom}} + 2$ mm. The red cone is the solid angle $\Omega(d)$ subtended by the crystal face of radius $r = 12.5$ mm. Right: Geant4 OGL cutaway view of the detector cut in half at $y = 0$, in the flush-source-to-detector-face geometry ($d_{\text{nom}} = 1$ mm, $d = 3$ mm); the source is at the left and the orange epoxy puck touches the detector front plane. The render uses opaque material colours: orange epoxy puck, grey aluminium casing, cream Teflon reflector, green CLYC crystal, yellow optical interface, pale blue PMT vacuum, dark PMT base, and dark grey MuMetal shield. The thin PMT glass envelope is omitted for clarity. The detector is 125 mm long. The left drawing is schematic; the distance scan at $d_{\text{nom}} = 200$ mm, 300 mm, and 400 mm is not drawn to scale.

For the geometries used here,

$$\epsilon_\Omega = \begin{cases} 0.0955\,\%, & d_{\text{nom}} = 200\,\text{mm}\ (d = 202\,\text{mm}), \\ 0.0428\,\%, & d_{\text{nom}} = 300\,\text{mm}\ (d = 302\,\text{mm}), \\ 0.0242\,\%, & d_{\text{nom}} = 400\,\text{mm}\ (d = 402\,\text{mm}), \\ 38.33\,\%, & d_{\text{nom}} = 1\,\text{mm}\ (d = 3\,\text{mm}) \quad \text{(flush)}. \end{cases} \tag{3}$$

1.2 Maximum efficiency bound and intrinsic efficiency

The total full-energy peak efficiency factorizes as

$$\epsilon = \epsilon_\Omega(d)\,\epsilon_{\text{int}}(E_\gamma, d), \tag{4}$$

where $\epsilon_{\text{int}} \in [0, 1]$ is the probability that a gamma entering the crystal deposits all of its energy. Because $\epsilon_{\text{int}} \leq 1$, the geometric acceptance is a hard upper bound,

$$\epsilon \leq \epsilon_{\max}(d) = \epsilon_\Omega(d). \tag{5}$$

2 Calculation Methods

2.1 Measured efficiency

The 2026-10-27 CLYC2 spectra are converted from CNF to ROOT with `cnf2root`. The converter writes the channel-count histogram, the calibrated energy histogram, and the acquisition metadata, including live time and the energy calibration. The analysis uses the native ~ 1.96 keV energy bins; rebinning to 2 keV by bin centre would alias the two grids.

The 600 s environmental background spectrum is subtracted bin by bin. This removes room and cosmic background; it does not remove source-induced Compton continuum. Each full-energy peak is then fitted with a Gaussian plus linear background over ± 1.5 FWHM around the observed peak. The fit range for the 1173.23 keV line is 1113.5 keV–1233.0 keV; for the 1332.51 keV line it is 1264.4 keV–1400.6 keV. The Gaussian area is the measured peak area.

The measured efficiency is

$$\epsilon_{\text{exp}} = \frac{N_{\text{peak}}}{A_{\text{exact}} P_\gamma t_{\text{live}}/100} \times 100\%, \tag{6}$$

where A_{exact} is the source activity at the measurement date, P_γ is the gamma-ray emission probability in percent, $t_{\text{live}} = 600\,\text{s}$, and N_{peak} is the Gaussian peak area. The statistical uncertainty is obtained from the Gaussian fit.

For the 1173.23 keV line, the peak sits on the Compton continuum from the 1332.51 keV line. A local sideband integration over the ±1 FWHM peak region with sidebands at 1.2–1.8 FWHM gives a peak area that differs from the Gaussian-plus-linear fit by about 10 %. This is treated as the background-model systematic for the 1173.23 keV line. For the 661.64 keV and 1332.51 keV lines the two methods agree to about 1 %.

2.2 Simulated efficiency

The simulations use `G4EmStandardPhysics_option4`. For each energy and distance, 10^7 monoenergetic gammas are generated isotropically from the puck centre. All energy deposited in the CLYC crystal is summed per event. Full-energy peak events satisfy

$$0.98E_\gamma \le E_{\text{dep}} \le 1.02E_\gamma. \tag{7}$$

The simulated efficiency and statistical uncertainty are

$$\epsilon_{\text{sim}} = \frac{N_{\text{peak}}}{N_{\text{inc}}} \times 100\%, \qquad \sigma_{\text{sim}} = \frac{\sqrt{N_{\text{peak}}}}{N_{\text{inc}}} \times 100\%, \tag{8}$$

with $N_{\text{inc}} = 10^7$. The simulation is not resolution-smeared here; the full-energy peak is a delta line, so the ±2% window contains the complete peak.

The simulation/measurement ratio and statistical pull are

$$R = \frac{\epsilon_{\text{sim}}}{\epsilon_{\text{exp}}}, \qquad \text{pull} = \frac{\epsilon_{\text{sim}} - \epsilon_{\text{exp}}}{\sqrt{\sigma_{\text{sim}}^2 + \sigma_{\text{exp}}^2}}. \tag{9}$$

2.3 Flush source response

For the flush geometry, the source puck touches the detector front face. The source is at the puck centre, 1 mm from the detector front plane and 3 mm from the crystal front face. The decay chain is tracked with `G4RadioactiveDecayPhysics` using parent ions at rest at the puck centre ($N_{\text{decays}} = 10^6$ per source). The long-decay threshold is 1×10^{60} y, and the analysis merges pulses whose leading-edge times differ by at most the 1 µs CLYC resolving time. The 1×10^4 y cross-check stops the ^{241}Am chain at ^{237}Np; it does not change the ^{137}Cs, ^{210}Pb, ^{152}Eu, or ^{60}Co results. The simulated peak efficiency is

$$\epsilon_{\text{sim}}(E_\gamma) = \frac{N_{\text{pulse}}(E_\gamma)}{N_{\text{decays}} P_\gamma(E_\gamma)} \times 100\%, \tag{10}$$

with the statistical uncertainty obtained from the pulse count.

2.4 The 2 cm calibration file

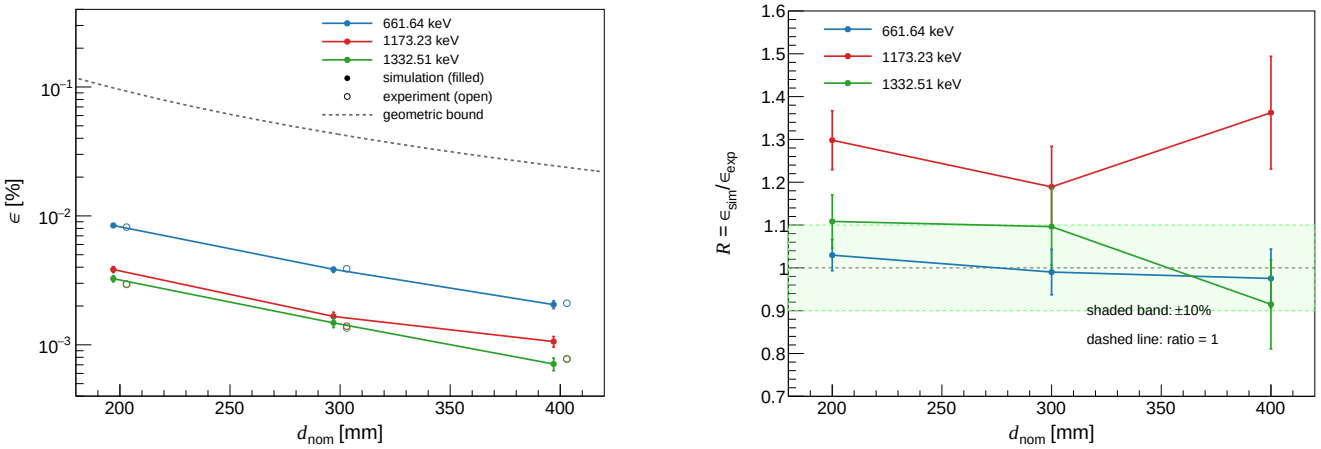
The calibration file `eff_clyc1.csv` is labelled as a 2 cm measurement. The file has no detector column and is treated as CLYC1-only. Its printed values match Eq. (6) without a factor of 1000. At lower energies (32 keV–121 keV) the values range from 15.4 % to 23.9 %. The file documents neither the actual source position nor the detector identity, so the measured values are compared with the flush simulation of Sec. 6.

3 Distance Comparison Results

Table 1 lists the nine CLYC2 points. The measured efficiencies are the Gaussian peak areas from the 2026-10-27 spectra; no external efficiency table is used. Figure 2 shows the efficiencies and the simulation/measurement ratios.

Table 1: Simulated and measured full-energy peak efficiencies for CLYC2 at the nominal source-to-detector distance d_{nom}. The source-to-crystal-front-face distance is $d = d_{\mathrm{nom}} + 2\,\mathrm{mm}$. The measured efficiency is from the Gaussian-plus-linear fit to the 2026-10-27 spectrum. All points satisfy $\epsilon < \epsilon_\Omega(d)$.

d_{nom} [mm]	E_γ [keV]	Source	ϵ_{sim} [%]	ϵ_{exp} [%]	R	Pull [σ]
200	661.64	^{137}Cs	0.008 43(29)	0.008 19(7)	1.030	+0.8
200	1173.23	^{60}Co	0.003 84(20)	0.002 96(3)	1.298	+4.4
200	1332.51	^{60}Co	0.003 26(18)	0.002 94(3)	1.108	+1.8
300	661.64	^{137}Cs	0.003 85(20)	0.003 89(5)	0.990	−0.2
300	1173.23	^{60}Co	0.001 66(13)	0.001 40(2)	1.189	+2.0
300	1332.51	^{60}Co	0.001 48(12)	0.001 35(2)	1.096	+1.1
400	661.64	^{137}Cs	0.002 05(14)	0.002 10(3)	0.975	−0.4
400	1173.23	^{60}Co	0.001 06(10)	0.000 78(2)	1.362	+2.8
400	1332.51	^{60}Co	0.000 71(8)	0.000 78(1)	0.915	−0.8



(a) Efficiency vs d_{nom} and the geometric bound.

(b) Simulation / measurement ratio.

Figure 2: Distance comparison for CLYC2. (a) Full-energy peak efficiency vs d_{nom}. Filled markers and solid lines are the simulation; open markers are the experimental Gaussian peak areas. The simulation and experiment markers are offset horizontally by ±3 mm so that they do not overlap; all quoted values and the table use the nominal d_{nom}. The dashed grey curve is the geometric bound from Eq. (2). (b) Ratio $R = \epsilon_{\mathrm{sim}}/\epsilon_{\mathrm{exp}}$. The green band is ±10 %; the dashed line is $R = 1$.

The ^{137}Cs 661.64 keV ratio is 0.98–1.03 at the three distances, with pulls between −0.4 and +0.8σ. The ^{60}Co 1332.51 keV ratio is 0.92–1.11, with pulls between −0.8 and +1.8σ. The 1173.23 keV ratio is 1.19–1.36, with pulls between +2.0 and +4.4σ.

The 1173.23 keV line is therefore the outlier. The measured 1173 keV/1332 keV ratio is 1.00–1.04, while the simulated ratio is 1.18. The 1173.23 keV Gaussian area has the 10 % background-model systematic from the sideband check; the 10 % background-model uncertainty does not remove the excess. The 1332.51 keV line changes by only 0.92–1.11, so the excess is line-dependent rather than a common normalization.

At 200 mm–400 mm the Co-60 cascade summing is negligible: the total interaction probability per gamma is about 3.3×10^{-4}, so the probability that both cascade gammas interact is about 10^{-7} and the 2505.74 keV sum peak is negligible. The 1173.23 keV excess is therefore not cascade summing. It points to an energy-dependent source or detector effect, such as the actual Co-60 source capsule or holder, the detector active volume, or the reflector/casing model. The available data do not identify which.

4 Experimental Spectra and Peak Integration

The 2026-10-27 CLYC2 spectra are the only measured spectra used in this report. The CNF conversion, 600 s background subtraction, and Gaussian fits are described in Sec. 3. The Gaussian peak areas for the nine points are the measured efficiencies in Table 1. The fit ranges are ±1.5 FWHM; the 1173.23 keV fit range is 1113.5 keV–1233.0 keV. The local sideband integration over the same peak region gives

an independent area; for 1173.23 keV it differs from the Gaussian fit by about 10 %, and this is the continuum-model systematic. For 661.64 keV and 1332.51 keV the two methods agree to about 1 %.

Figure 6 shows the Gaussian-plus-quadratic-background fits to the 200 mm Co-60 spectra. This figure is a diagnostic: the table uses the Gaussian-plus-linear fit for the measured efficiency, while the figure uses a quadratic background because the 1173.23 keV peak sits on the curved 1332.51 keV Compton continuum. The fitted simulation/experiment peak-area ratios in the figure are 1.21 at 1173.23 keV and 1.04 at 1332.51 keV; the table gives 1.30 and 1.11. The difference between the two ratios is the background-model uncertainty of the 1173.23 keV line.

5 Detector Resolution and Simulated Spectrum Comparison

5.1 Resolution model

The FWHM of the 661.64 keV, 1173.23 keV, and 1332.51 keV peaks is extracted from the same Gaussian-plus-linear fits at 200, 300, and 400 mm. The three distances give consistent widths; Table 2 lists the mean and the standard error of the mean. The resolution $R = \mathrm{FWHM}/E$ is 4.80 % at 661.64 keV, 3.40 % at 1173.23 keV, and 3.41 % at 1332.51 keV. The 1173.23 keV width is affected by the broad background under the peak, so it is not used in the fit. The two stable lines give

$$\mathrm{FWHM}(E) = 1.239\sqrt{\mathrm{E/keV}}\,\mathrm{keV}, \qquad R(E) = \frac{123.9 \pm 0.4}{\sqrt{E/\mathrm{keV}}}\,\%.$$

The 1173.23 keV point lies 6.2 % below this curve, consistent with the background-model bias.

Table 2: Measured FWHM and resolution of the three CLYC2 lines, averaged over the 200, 300, and 400 mm spectra. The uncertainty is the standard error of the mean over the three distances. The model is $R(E) = 123.9/\sqrt{E/\mathrm{keV}}\,\%$, fitted to the 661.64 keV and 1332.51 keV lines; the 1173.23 keV point is a cross-check.

E_γ [keV]	n	FWHM [keV]	R [%]	R_{model} [%]	residual [%]	fit
661.64	3	31.786(86)	4.804(13)	4.818	−0.3	yes
1173.23	3	39.837(183)	3.396(16)	3.618	−6.2	no
1332.51	3	45.372(550)	3.405(41)	3.395	+0.3	yes

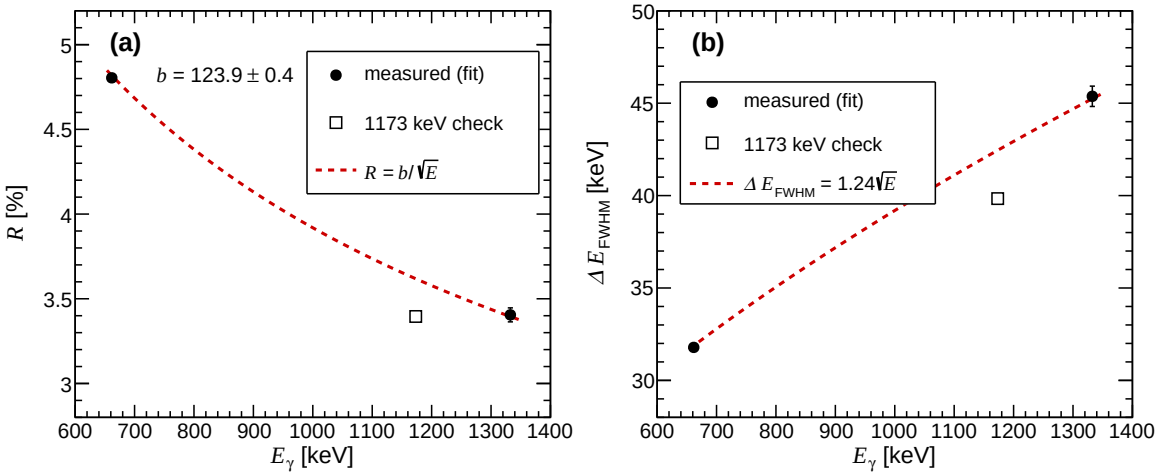


Figure 3: (a) Resolution $R = \mathrm{FWHM}/E$ and (b) FWHM versus E_γ. Filled circles are the 661.64 keV and 1332.51 keV lines used in the fit; the open square is the 1173.23 keV cross-check. The dashed curve is $R(E) = 123.9/\sqrt{E/\mathrm{keV}}\,\%$ ($\mathrm{FWHM} = 1.239\sqrt{E/\mathrm{keV}}\,\mathrm{keV}$).

The model is used only to smear the simulated spectra: each event with deposited energy E_{dep} is replaced by a Gaussian draw with mean E_{dep} and standard deviation $\sigma(E_{\mathrm{dep}}) = \mathrm{FWHM}(E_{\mathrm{dep}})/2.3548$. The smearing conserves the number of events and does not change the peak areas. The model is an extrapolation between and below the fitted lines.

5.2 Simulated and experimental spectra at 200 mm

The 200 mm simulated spectra use the existing 10^7-event monoenergetic runs, with the GPS emission restricted to a 15 degree cone around the detector axis. The cone increases the number of events that reach the crystal by the inverse of the cone solid-angle fraction $f = (1 - \cos 15^\circ)/2 = 0.017037$; the equivalent isotropic incident count is $N_{\mathrm{inc,eff}} = N_{\mathrm{gen}}/f = 5.87 \times 10^8$. The cone run reproduces the isotropic peak efficiencies within about 1σ: 0.008 71 % versus 0.008 43 % at 661.64 keV, 0.003 73 % versus 0.003 84 % at 1173.23 keV, and 0.003 08 % versus 0.003 26 % at 1332.51 keV. The cone cut removes gammas emitted outside 15 degrees that scatter into the crystal; those gammas contribute mainly to the continuum, so the peak comparison is quantitative and the continuum comparison is qualitative.

Both simulated spectra (with and without resolution) use the same events and the same scaling. The number of emitted gammas is

$$N_{\mathrm{emit}} = A_{\mathrm{exact}} P_\gamma t_{\mathrm{live}}/100,$$

and the scale factor is $N_{\mathrm{emit}} f/N_{\mathrm{gen}}$. For Cs-137 at 200 mm, $N_{\mathrm{emit}} = 2.834 \times 10^8$ and the scale factor is 0.4828. For Co-60, the 1173.23 keV and 1332.51 keV runs are summed, each with its own N_{emit} (7.031×10^8 and 7.032×10^8) and scale factor 1.198. The unsmeared spectrum fills each event at its deposited energy; the smeared spectrum replaces it by a Gaussian draw with $\sigma(E_{\mathrm{dep}}) = \mathrm{FWHM}(E_{\mathrm{dep}})/2.3548$. The 2505.74 keV sum peak is not present in the monoenergetic runs.

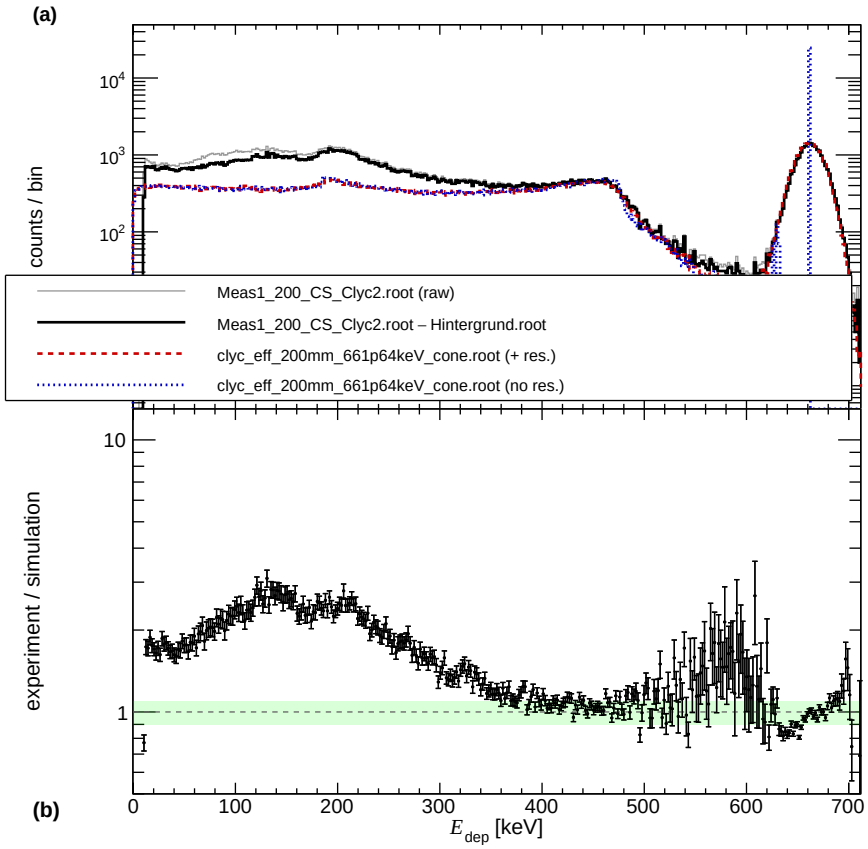


Figure 4: Simulated and experimental Cs-137 spectra at $d_{\mathrm{nom}} = 200$ mm. (a) Raw experimental spectrum (grey), 600 s background-subtracted spectrum (black), resolution-smeared simulation (red dashed), and simulation without resolution (blue dotted), all scaled to the measured live time and activity. (b) Ratio of the background-subtracted spectrum to the resolution-smeared simulation over the full x-range; the ratio axis is logarithmic, the dashed line is ratio = 1, and the green band is ±10%. The simulation uses the 15 degree cone run; the 661.64 keV peak is quantitative, the low-energy continuum is qualitative because the simulation contains only the 661.64 keV gamma. The experimental file is `Meas1_200_CS_Clyc2.root`, the background file is `Hintergrund.root`, and the simulation file is `clyc_eff_200mm_661p64keV_cone.root`.

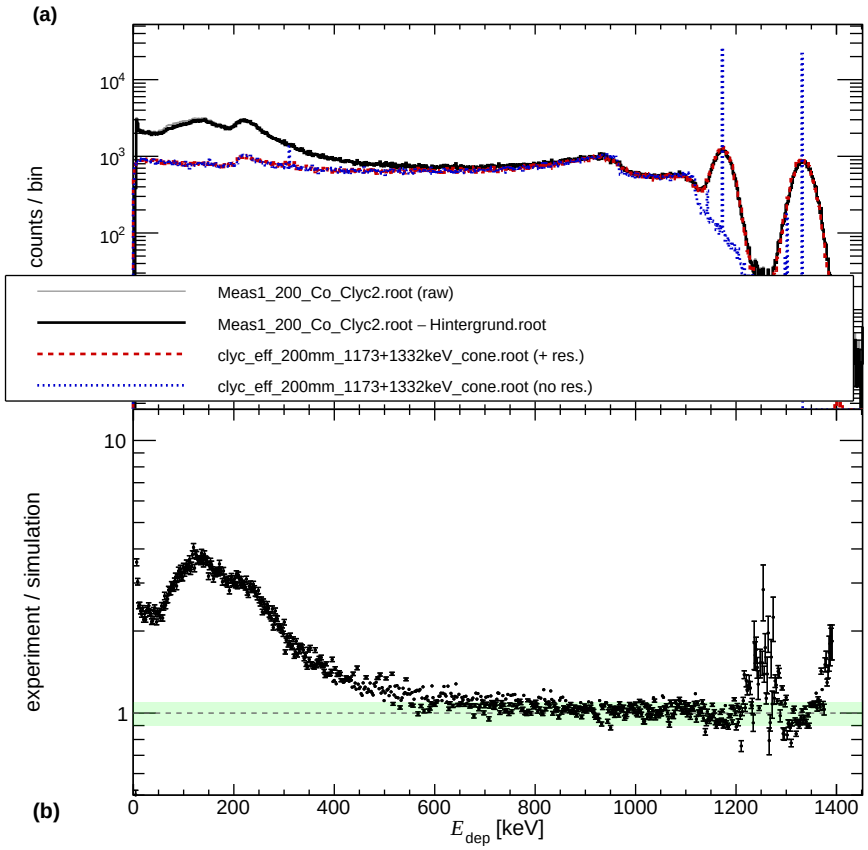


Figure 5: Simulated and experimental Co-60 spectra at $d_{\mathrm{nom}} = 200\,\mathrm{mm}$. (a) Raw experimental spectrum (grey), 600 s background-subtracted spectrum (black), and the sum of the 1173.23 keV and 1332.51 keV simulations with resolution (red dashed) and without resolution (blue dotted), all scaled to the measured live time and activity. (b) Ratio of the background-subtracted spectrum to the resolution-smeared simulation over the full x-range; the ratio axis is logarithmic, the dashed line is ratio = 1, and the green band is ±10%. The 2505.74 keV sum peak is absent from the monoenergetic simulation. The ratio in (b) is a bin-by-bin interval comparison; at 1173.23 keV the interval contains Compton continuum from the 1332.51 keV line, so it is not the full-energy peak efficiency of Table 1. The experimental file is `Meas1_200_Co_Clyc2.root`, the background file is `Hintergrund.root`, and the simulation is the sum of `clyc_eff_200mm_1173p23keV_cone.root` and `clyc_eff_200mm_1332p51keV_cone.root`.

For Cs-137 at 200 mm, the resolution-smeared simulation reproduces the 661.64 keV peak: the ratio of the background-subtracted counts to the smeared simulation in the ±2% interval is 0.955, and the smeared simulated peak area is about 5 % above the measured Gaussian area. The simulation lies below the measured continuum at low energy; the experimental spectrum contains room background and scattered radiation from the source environment, which the monoenergetic simulation does not. The 200 mm Cs-137 spectrum has no resolved 32 keV Ba X-ray peak, so the source encapsulation may absorb the X-rays; the simulation does not model the encapsulation. The 661.64 keV peak itself is the quantitative comparison.

For Co-60 at 200 mm, the table gives $\epsilon_{\mathrm{sim}}/\epsilon_{\mathrm{exp}} = 1.30$ at 1173.23 keV and 1.11 at 1332.51 keV. The bin-by-bin ratio in Fig. 5(b) is not the same quantity. The ±2% interval at 1173.23 keV is 46.9 keV wide, or 1.18 FWHM, and contains Compton continuum from the 1332.51 keV line: the background-subtracted interval integral is 23 396 counts, compared with 20 800 in the Gaussian peak, while the smeared simulation gives 24 306 counts in the interval and about 26 200 in the full-energy peak. The interval ratio is therefore 0.963, whereas the full-energy peak ratio is 20,819/26,210 = 0.79 (simulation/measurement 1.26), consistent with the table ratio 1.30. Figure 6 shows the fitted peaks and their Gaussian and background components; the fitted ratios are 1.21 and 1.04. The 1173.23 keV peak area has the 10 % background-model uncertainty from Sec. 4; the excess is larger. The 1332.51 keV agreement is within the peak-integral uncertainty. The experimental Co-60 spectrum has lines at 141.8 keV and 222.1 keV that are not Co-60 lines and are absent from the simulation. The 2505.74 keV sum peak is also absent; at 200 mm it is expected to be small, about 5 counts, so it does not explain the 1173.23 keV excess. At 200–400 mm the total interaction probability per gamma is about 3.3×10^{-4}, so the summing-out probability is below 0.05 %. The Co-60 excess is therefore not cascade summing; the Co-60 source geometry

or the detector response model is the more likely cause.

The conversion, analysis, and plotting scripts are `report/eff_clyc/scripts/convert_exp_spectra.sh`, `analyze_exp_spectra.C`, `make_exp_sim_figs.C`, and `make_distance_figs.C`. The peak-integration summaries are in `report/eff_clyc/data/`; the 15 degree cone runs are produced by `scripts/run_clyc_eff_cone_200mm.sh`.

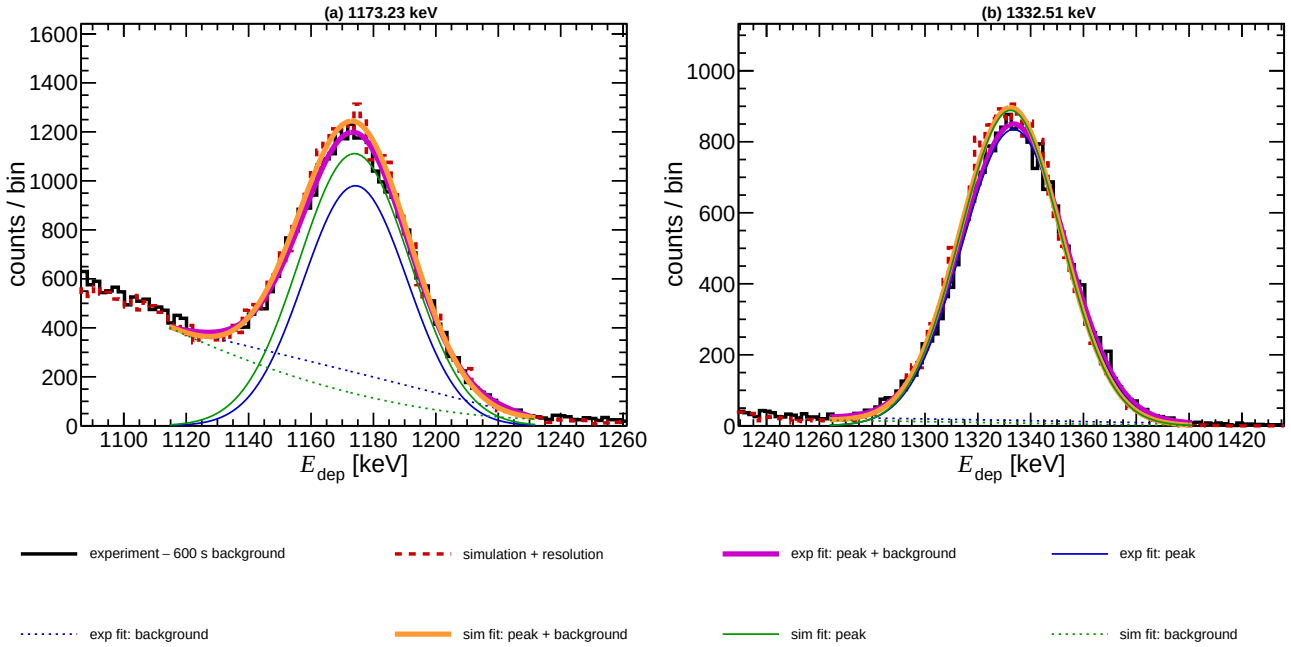


Figure 6: Gaussian-plus-background fits to the 200 mm Co-60 spectra. (a) 1173.23 keV and (b) 1332.51 keV. Black: experiment after the 600 s environmental background subtraction. Red dashed: resolution-smeared simulation. Both spectra are fitted over ±1.5 FWHM of the measured peak with a Gaussian plus quadratic background. For each fit, the thick magenta/orange curve is the peak-plus-background fit, the thin blue/green curve is the Gaussian peak, and the dotted blue/green curve is the background. The fitted peak areas are 20 592 (experiment) and 24 885 (simulation) at 1173.23 keV, and 20 670 and 21 438 at 1332.51 keV; the simulation/experiment ratios are 1.21 and 1.04. The quadratic background is used because the 1173.23 keV peak sits on the curved Compton continuum from the 1332.51 keV line. The data files are `Meas1_200_Co_Clyc2.root` and `Hintergrund.root`; the simulation is the sum of `clyc_eff_200mm_1173p23keV_cone.root` and `clyc_eff_200mm_1332p51keV_cone.root`.

6 Radioactive-Decay Source Response in the Flush Geometry

Table 3 lists the simulated peak efficiencies for the nine lines in the flush-source-to-detector-face geometry ($d_{\mathrm{nom}} = 1$ mm, $d = 3$ mm; $N_{\mathrm{decays}} = 10^6$ per source). The file has no detector column and is treated as CLYC1-only; the simulated curve is the same for both detectors because the geometry is identical. Figure 7 plots the flush simulation alongside the printed CSV values and the geometric acceptance; panel (b) shows the intrinsic efficiencies.

Table 3: Simulated peak efficiencies in the flush-source-to-detector-face geometry ($d_{\mathrm{nom}} = 1$ mm, $d = 3$ mm; $N_{\mathrm{decays}} = 10^6$ per source). The efficiency is the number of 1 µs pulses in the ±2 % window divided by $N_{\mathrm{decays}} P_\gamma$. The source runs use the 1×10^{60} y threshold; the ^{241}Am 1×10^{4} y cross-check is discussed in the text. The last column is $R = \epsilon_{\mathrm{flush}}/\epsilon_{\mathrm{CSV}}$.

Source	E_γ [keV]	$\epsilon_{\mathrm{flush}}$ [%]	ϵ_{CSV} [%]	R
^{137}Cs	32.00	15.900	18.50	0.859
^{210}Pb	46.54	25.233	23.85	1.058
^{241}Am	59.54	28.614	23.80	1.202
^{152}Eu	121.78	18.006	15.40	1.169
^{152}Eu	344.28	6.274	4.87	1.288
^{137}Cs	661.64	2.057	1.87	1.100
^{60}Co	1173.23	0.892	0.51	1.749
^{60}Co	1332.51	0.706	0.56	1.261
^{152}Eu	1408.01	0.431	0.40	1.079

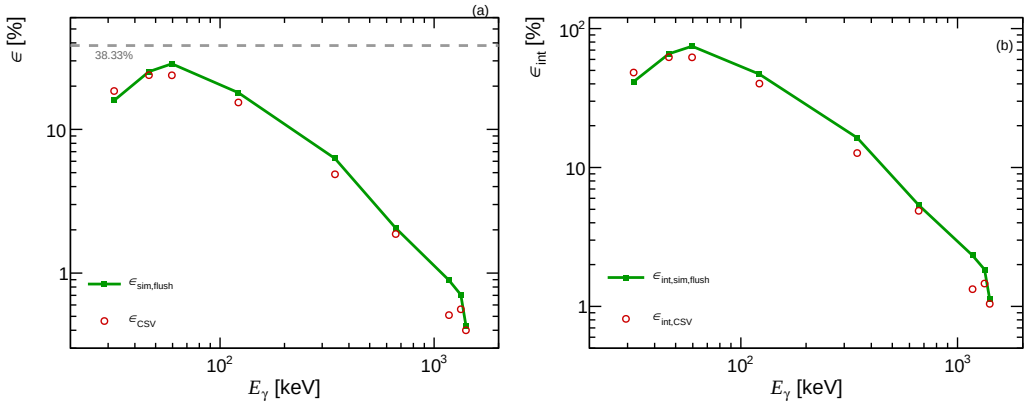


Figure 7: Simulated ϵ in the flush-source-to-detector-face geometry compared with the printed CSV values. (a) Raw efficiency; the dashed line marks the geometric acceptance $\epsilon_\Omega(3\,\mathrm{mm}) = 38.33\,\%$. (b) Intrinsic efficiency; the flush simulation and the CSV are divided by $\epsilon_\Omega(3\,\mathrm{mm})$.

The flush simulation is physically consistent throughout: it stays below the 38.33 % geometric bound at every line, and the 32 keV peak efficiency (15.90 %) is lower than the 46.54 keV and 59.54 keV values because photoelectric absorption in the 2 mm epoxy puck attenuates the 32 keV X-ray. Above 122 keV, the efficiency drops smoothly with energy to 0.431 % at 1408 keV.

A puck touching the detector front face puts the source 1 mm from the front plane and 3 mm from the crystal. The puck is 2 mm thick and the source is at its centre; the crystal front face is 2 mm behind the front plane. Eq. (2) then gives $\epsilon_\Omega(3\,\mathrm{mm}) = 38.33\,\%$. Figure 7 compares the raw and intrinsic efficiencies of the flush simulation with the printed CSV values; Table 3 lists the numbers. The flush simulation agrees with the CSV at 46.54 keV, 661.64 keV, 1408.01 keV, and 59.54 keV after the 1 µs pulse clustering ($R = 1.058$, 1.100, 1.079, and 1.202); the 32.00 keV line is 15.90 % against 18.50 % ($R = 0.859$), and the ^{60}Co lines remain high ($R = 1.749$ and 1.261).

The 59.54 keV value changed most when the analysis moved from an event sum to pulse counting. Summing all hits in an event gives 11.41 % in the flush geometry. The `hits` tree stores one row per shower pulse; merging rows whose leading-edge times differ by at most 1 µs gives 28.61 %. The 1×10^4 y cross-check gives 28.06 %; the 0.56-percentage-point difference is the late-chain contribution. In the 1×10^{60} y file, 2204 of the 101 387 rows in the 59.54 keV window have $t \geq 1 \times 10^{21}$ ns, so those pulses come from ^{237}Np or its descendants rather than from the primary decay. They are separate in time from the prompt pulse, so the clustering does not remove them; it prevents them from being summed with it. The 1×10^{60} y run is a full-chain stress test; for a fresh ^{241}Am source the 1×10^4 y result is the physical one.

The ^{60}Co lines remain high after the pulse clustering. This is not a missing-cascade effect: the flush ^{60}Co simulation includes the cascade, so the 1173.23 keV and 1332.51 keV peaks already include coincidence summing and the 2505.74 keV sum peak is present. At 3 mm the summing-out is strong, but the simulation still gives 0.892 % and 0.706 % against 0.51 % and 0.56 % in the CSV. The flush hypothesis is plausible for some lines and not for others; no parameter was tuned.

7 Conclusions

1. The geometric solid angle sets a strict physical ceiling $\epsilon \leq \epsilon_\Omega(d)$, with upper bounds of 0.0955 % at 200 mm, 0.0428 % at 300 mm, 0.0242 % at 400 mm, and 38.33 % for the flush-source-to-detector-face geometry ($d = 3$ mm).
2. The 661.64 keV simulation/measurement ratio is 0.98–1.03, with pulls between -0.4 and $+0.8\sigma$.
3. The 1332.51 keV simulation/measurement ratio is 0.92–1.11, with pulls between -0.8 and $+1.8\sigma$. The 1173.23 keV ratio is 1.19–1.36, with pulls between $+2.0$ and $+4.4\sigma$. The measured 1173 keV/1332 keV ratio is 1.00–1.04, compared with 1.18 in the simulation. The line-dependent excess is not cascade summing at 200–400 mm; the Co-60 source geometry or the detector response model is the more likely cause.

4. The distance-comparison efficiencies are from Gaussian-plus-linear fits to the 2026-10-27 CLYC2 spectra. The 1173.23 keV Gaussian area has a 10 % background-model systematic from the sideband check; the 661.64 keV and 1332.51 keV areas agree with the sideband method to about 1 %. No external efficiency table is used for the distance comparison.
5. The flush-source simulation agrees with the 2 cm calibration file at 46.54 keV, 661.64 keV, 1408.01 keV, and, after the 1 µs pulse clustering, 59.54 keV; the 121.78 keV, 344.28 keV and ^{60}Co lines remain high, so source position alone does not explain the scale difference.
6. The detector resolution is $R(E) = (123.9 \pm 0.4)/\sqrt{E/\mathrm{keV}}\,\%$ (FWHM $= 1.239\sqrt{E/\mathrm{keV}}\,\mathrm{keV}$) over 661 keV–1332 keV. The resolution-smeared 200 mm simulation reproduces the 661.64 keV peak within about 5 % and the 1332.51 keV peak within about 5 %; the 1173.23 keV full-energy peak area is 19–36% larger than the measurement.